Verify, Look Again, Remember: Fail-Closed Vision-Language Checks and Safety-Monotone Memory for Relief Delivery Robots

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Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

***Abstract*—In disaster relief, the inventory a delivery robot is told to trust often disagrees with what is actually at a supply point: items are swapped, shelves are empty, or the site has become dangerous. We present RescueMind, a mobile robot that checks the item and the site with a vision-language model (VLM) on a local GPU before it carries anything to a shelter. The check is built to fail closed: decisions come from the next-token probabilities of answer letters in closed-set questions, never from free text, and motion stays behind a velocity guard that the VLM cannot command. Running the full stack exposed two gaps, and we close each with a simple, verifiable rule. First, fires partly hidden behind the supply escape a single frontal view, so the robot re-views the station from 0.4 m to either side before it leaves. In a paired, pre-registered campaign this reduced deliveries despite a perceivable fire from 11 to 1 of 16 (exact McNemar $p = 0.0019$), with no false stops. Second, a robot that resupplies the same shelter can remember what it saw, but stale memory is dangerous. We let memory only reorder where the robot looks and add caution, never remove a check, which bounds the per-mission probability of an unsafe delivery by the number of stations checked times the per-check false-approval rate, however stale the memory is. In 32 paired missions per arm in worlds that change between missions, unsafe deliveries were 0 with our memory, 1 without memory and 20 with a naive memory that skips checks; where supplies stayed relocated, the median mission time fell from 114 s to 79 s. The held-out tests and closed-loop campaigns were pre-registered, and every closed-loop trial was audited against simulator ground truth: 60/60 expected outcomes in randomised worlds, 0/2304 false approvals under 12 camera degradations, against 44/408 for a zero-shot SigLIP baseline on held-out images. On real photographs the hazard check found 298/300 fire or smoke images but also flagged 64/100 fire-free ones, so a site-specific threshold is needed before deployment. We release code, worlds, ledgers and every amendment.**

I. Introduction

Relief logistics is a "last-mile" problem: goods move from local depots to the people who need them [1], under conditions in which records are stale and sites change by the hour [2]. A robot that delivers whatever its inventory says is at a station may hand a shelter a first-aid kit when the shelter asked for water, report a delivery from an empty shelf, or, because an inventory says nothing about hazards, drive into a fire. Vision-language models (VLMs) can often describe such situations without task-specific training and are used to explain or detect robot failures [3]–[5]. Their outputs, however, are biased by option order and answer letters [6], [7], they hallucinate objects that are not there [8], and free-form generation can be unparseable. We saw all three on our own images (Supp. Sec. S8).

Relief robots also return to the same shelter again and again, and a robot that remembers where the water really was, or where it saw smoke, could be faster. Yet acting on stale memory can hurt: in a FrozenLake-style grid world with six VLMs, an agent that trusted unaudited memory failed more often than one with no memory [9], and a disaster site changes between missions. So, *how can a robot benefit from memory without letting memory weaken its checks?*

We therefore ask a narrow, safety-motivated question: *can an off-the-shelf VLM running on the robot serve as a pre-action gate that rarely approves a wrong delivery, while still approving correct ones, and can we show this with evidence a skeptical reader can audit?* Our answers are a system design and an evaluation methodology:

- **A fail-closed checkpoint** (Sec. III-A). Two permuted multiple-choice questions and a Yes/No hazard question are scored by answer-letter probabilities; the robot approves only if they agree and clear a confidence threshold, as in a selective classifier [10]. A wrong item triggers a one-time search of the other station, a hazard never does.
- **Active look-around** (Sec. V-C). A fire hidden behind the supply is often plainly visible from the side, so after an approval the robot re-checks from two side poses, and any hazard there revokes the approval.
- **Safety-monotone mission memory** (Sec. III-D). Memory proposes, perception disposes: it may reorder the stations the robot visits and tighten its hazard threshold, but never skip a check. We state the invariant, enforce it in code and in the audit, and test it in a pre-registered multi-mission campaign against a naive memory.
- **The original two-model design on one GPU** (Sec. III-C). We fill the second checker's role with Clef-Flash, which Cloudflare post-trained from Qwen3.5-9B [11], require both to approve, and run the pair on the simulated robot.
- **Decision and motion separated, in the spirit of Simplex** [12], [13]: the mission node only *requests* velocities,

and an independent guard is the sole publisher of wheel commands.

- **Auditable, pre-registered evaluation** (Sec. IV). Prompts and thresholds were chosen on development images only; a 600-image held-out test set and the closed-loop campaigns D–J and V were registered before running [14], [15]. Each closed-loop trial is checked against Gazebo ground truth, a ROS bag and a recomputation of the decision, and every attempt is kept. The studies in Table I include a camera-degradation study.
- **Lessons from running the real stack** (Supp. Sec. S8): 23 failure causes that unit tests and mocks missed, such as a wheel-axis frame error that let odometry-based audits pass a robot that never arrived.

Terms. *v3.4* is the first registered checkpoint (hazard threshold $\tau_h = 0.5$); *v3.5* lowers it to 0.05; *v3.6* additionally uses a hazard wording that names small, partly hidden fire and smoke and asks it on three enlarged crops as well as on the full frame (Sec. V-C). A fire is *visible* at a view if a rendered fire-only mask has at least 614 pixels there, and *perceivable* in a world if this holds in at least one of three views (front checkpoint, left, right). A *campaign* is a closed-loop study. *Expected behaviour* is the action the generator's ground truth requires: deliver, deliver after searching the other station, or refuse because of a wrong item or a hazard.

II. Related Work

Failure detection and verification. REFLECT [3], AHA [4] and success detectors [16] explain or detect failures after the fact; Sentinel [5] and FAIL-Detect [17] monitor learned policies at run time, Code-as-Monitor [18] checks constraints with VLM-generated programs, and an LLM anomaly monitor triggers a safe fallback [19] (Supp. Sec. S3). The closest work we know is VAP-TAMP [20]: before and after each action it asks a VLM paraphrased yes/no questions, takes a majority vote, and moves the camera if the view is insufficient; the vote updates the state used for replanning. To our knowledge, ours differs in closed-set next-token scores, a fail-closed conjunction (a failed check can only refuse or revoke), an independent velocity guard, memory limited to reordering and caution, and pre-registered closed-loop audits against ground truth. KnowNo [21] scores multiple-choice answers by next-token probability and asks a human for help with a conformal guarantee [22]; bounding a miss rate with high probability is risk control [23], [24]; SayCan [25] likewise scores a fixed candidate set. We share the scoring but refuse or search under uncertainty, and add permuted-option agreement against letter bias [6], [7], [26]. Language models are fairly calibrated on formatted multiple-choice questions [27], but confidence still needs validation [28], [29], and embodied VLMs abstain poorly on their own [30].

Safety architectures. Simplex [12], [13], shields [31] and control barrier functions [32] keep a simple verified layer between a complex decision maker and the actuators (see [33]). LLM-controlled robots can be manipulated into harmful actions [34]; RoboGuard [35] checks plans first. Unlike vision-language-action models [36], [37], our VLM never commands motion.

Memory and active perception. Semantic memory [38], [39] and 3D scene graphs [40], [41] *answer* or *plan* [42]; mapping has long dealt with change [43], [44]. In a grid-world study, unaudited stale memory was worse than none [9]; GLOVE [45] and VLM-DEWM [46] check memory against fresh observations. We instead use memory to decide *where and how carefully to verify*, never to reduce verification. Object search with belief priors [47]–[50] and semi-static maps [51], [52] also use past observations to order search; to our knowledge they trust detections and do not bound the risk of stale memory at a pre-action gate. Moving the sensor to resolve ambiguity is classic [53], [54], and agents explore until a VLM is confident [55]; ours is deliberately simple: two fixed side views, used only to *revoke* an approval.

Robustness, redundancy and evaluation. Classifiers, detectors and VLMs degrade under common corruptions [56]–[59]; we measure one part of the sim-to-real gap [60], [61], sensor degradation, and care about the *direction* of errors. Agreement between independent versions is a classic fault-tolerance technique [62] limited by correlated failures [63]. We follow recommendations to report conditions, metrics, uncertainty and failure modes [64], with Wilson intervals [65], exact McNemar tests [66] and domain randomisation [67] for the held-out *test* set.

III. System

Platform. A differential-drive robot with a 2D lidar and a mast camera (640×480, 1.05 rad horizontal field of view) runs in Gazebo Harmonic with ROS 2 Jazzy [68], [69] (Fig. 1). The arena has two supply stations (water, medicine) and a shelter. The VLM runs in FP16 on a single Tesla V100 (32 GB). We use Qwen3.5-9B [70] (Apache-2.0), selected over Qwen2.5-VL-3B [71], Qwen3-VL-8B [72] and Qwen3.5-4B on development data (Sec. IV).

Mission. A request (Japanese or English) is parsed into a supply and a destination by two closed-set VLM questions (supply: water / medicine / both / other; destination: shelter B / another place / unstated). The robot accepts the parse only if the most likely item is water or medicine with $p \geq 0.6$, the most likely destination is not "another place" (shelter B is the only shelter), and an explicit rule finds no other shelter name; otherwise it refuses before moving. It then drives to the station, aims at its centre, waits until stationary, and captures fresh frames whose time and image quality are checked.

A. Fail-closed checkpoint

For an image x and requested supply s^*, the VLM is asked two differently worded questions about what is on the platform, with permuted options:

- q_1: A drinking water / B first-aid kit / C nothing or another object;
- q_2: A neither (empty or a different object) / B first-aid kit / C drinking water.

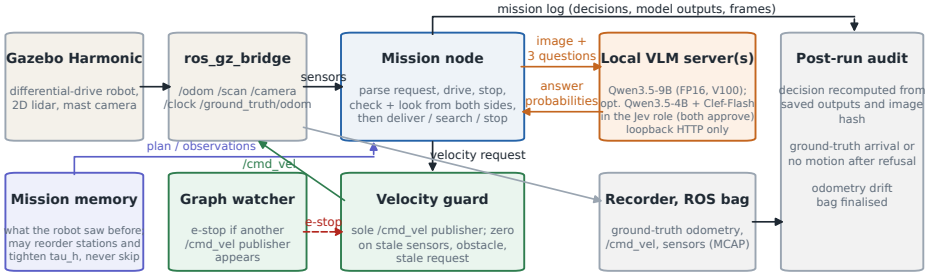


Fig. 1. Processes and data flow. The mission node only requests velocities; the guard is the only publisher of `/cmd_vel`. The VLM server is local; no cloud API is used. Ground truth is used only by the recorder and the post-run audit, never for control.

It is also asked (once per image) the hazard question q_h: "Look at the whole image, including behind and around the objects. Do you see any fire, flames, or smoke? Yes/No". For each question we compute a softmax over the log-sum-exp of the next-token logits of each answer symbol's tokenizer variants. This gives $p_k(s)$ for $k \in \{1, 2\}$ and p_h. The robot approves iff

$$\bigwedge_{k=1,2} \left[\arg\max_s p_k(s) = s^* \ \wedge \ p_k(s^*) \geq \tau_s \right] \ \wedge \ p_h < \tau_h, \quad (1)$$

with $\tau_s = 0.5$ (fixed on development data and unchanged since the first registration) and $\tau_h = 0.05$. In v3.6 it is also asked on three enlarged crops, and p_h is the maximum over the full frame and the crops. We call p_k and p_h probabilities for brevity: they are the next-token mass of the answer letters, renormalised over the allowed answers, and are scores to be thresholded, not calibrated probabilities (on real photographs their scale shifts; Sec. V). The hazard threshold was 0.5 up to campaign D and was lowered once (v3.5) after a pilot missed a mostly occluded fire (Supp. Sec. S8). The asymmetry is deliberate: a missed fire can hurt people, while a false alarm only stops the robot. Every failure leads to refusal: inference errors, non-finite logits (FP16 overflow guard), unknown labels, images with different hashes, or a model not on the allow-list. A refusal is followed by a 2.5 s zero-velocity hold that the audit checks.

Recovery and hazard policy. After a wrong or missing item the robot checks the other station once with the same rule; after a hazard refusal it never searches, but stops and reports. After every mission a one-page operator report is generated from the robot's own log: every check image with its decision and probabilities, the memory's proposals and changes, and a concrete next action (e.g. "ask the fire service to check station M", "update the inventory: the medicine is at station W").

Velocity guard. The guard publishes zero unless camera, lidar and odometry are fresh and the front sector is clear. A stale *motion* request latches the guard and ends the mission. A stale *zero* request holds zero without latching (Supp. Sec. S8). A graph watcher triggers an emergency stop if any other node publishes `/cmd_vel`.

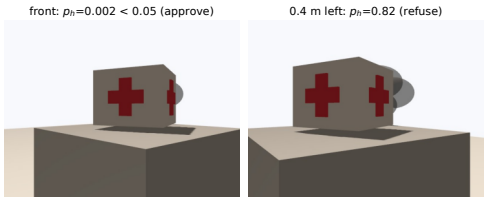


Fig. 2. Why the robot looks again. The same small fire, from the front checkpoint and from 0.4 m to the left (a v3.6 mission in a campaign-G world). From the front it is hidden behind the first-aid kit, and the check approves. From the side, the hazard question refuses.

B. Active look-around

A fire behind the supply is occluded by the very object the robot is checking. A larger model cannot see pixels that are not in the image, so the robot moves instead of thinking harder, in the spirit of active perception [53], [54]. After the front check approves, the robot drives to two poses 0.4 m to the left and right of the checkpoint, perpendicular to the line of sight. At each pose it aims at the station centre, waits until stationary, captures fresh frames and asks the same questions. Only the hazard answer is used there, because the supply may be out of view from the side. Any side view with a hazard turns the approval into a hazard refusal, with no search (Fig. 2). Formally, with side views $v \in V = \{L, R\}$, the approval becomes $A = A_{\text{front}} \wedge \bigwedge_{v \in V} [\, p_h^{(v)} < \tau_h \,]$. The audit recomputes every side decision. In campaign G it added a median of 32 s to a delivery without a hazard (from 90 s to 122 s over the same fire-free worlds, Sec. V-C).

C. A second checker

The original design of this robot asked two models (a second checker, called Jev in that design, and Qwen) and moved only when both agreed, following the idea of diverse redundancy [62], although failures are known to be correlated in practice [63]. The second checker is Clef-Flash, a decision model released by Cloudflare [11], [73] and served locally through the original second-checker API. According to its model card it is post-trained from Qwen3.5-9B, so the two models share lineage. Their diversity lies in post-training, in model size (our on-robot pair uses Qwen3.5-4B) and in the question asked, and the correlated-failure caveat applies in full. The two-model rule approves only if Eq. (1) holds for Qwen

and Clef-Flash agrees (thresholds: Supp. Sec. S1). A hazard seen by either model stops the robot, and a wrong supply seen by either allows the one-time search. Whether the two models fit one GPU is tested in Sec. V-F. The second model is a veto whose errors overlap with Qwen's. Qwen misses 17 of 211 development fires, and Clef-Flash also misses 15 of these (88%, Wilson [66, 97]%; Supp. Table S9).

D. Safety-monotone mission memory

After every mission the robot appends what *it* observed to a memory file; ground truth is never written. For each checked station the record holds the supply both passes named with $p \geq 0.5$ (or *uncertain*), whether a hazard was seen, and from which views (front, left, right). Before the next mission, memory may change exactly two things. (i) *Route proposal:* the robot may visit first the station where it last saw the requested supply, when the supply's own station last showed something else; that station is still checked, and the one-time search still applies. (ii) *Caution:* at a station where a hazard was seen during the last three missions, every view refuses at $p_h \geq \tau_h^{\text{mem}} = 0.01$ instead of $\tau_h = 0.05$, and the side that revealed the hazard is looked at first.

Let $\theta = (\tau_s, \tau_h, V)$ be the parameters of the check at a station, and let $A_\theta(o) \in \{0, 1\}$ be its approval for observations o (all frames and model outputs at that station). Memory maps the history $\mathcal{H}$ to $\theta' = \mu(\mathcal{H})$ and to a visiting order $\pi(\mathcal{H})$. We call a memory policy *verification-preserving* if, at every station it checks, $\tau'_s \geq \tau_s$, $\tau'_h \leq \tau_h$ and $V' \supseteq V$. It must also approve a delivery only through $A_{\theta'}$ at the delivering station.

Observation 1 (per-check monotonicity). For a verification-preserving policy, $A_{\theta'}(o) \leq A_\theta(o)$ for every o. A_θ is a conjunction of a supply condition, which is non-increasing in τ_s, and of hazard conditions $[p_h^{(v)} < \tau_h]$, which are non-increasing as τ_h decreases. Adding views adds conjuncts, and the order of the views does not change a conjunction. It is elementary by design, so code and audit can enforce it.

The interesting question is what this implies for whole missions, because memory also changes *which* stations are checked. Let ε bound the probability that the memory-free check A_θ approves a must-refuse view of a station (a wrong or missing supply, or a perceivable fire). The bound must hold for every station view the robot can encounter.

Corollary 1 (per-mission risk). A verification-preserving policy delivers only after $A_{\theta'}$, and hence A_θ, approved the delivering station. A mission checks at most k stations ($k = 2$ here: the first station and the one-time search). An unsafe delivery therefore requires a false approval at one of at most k must-refuse checks, and by the union bound $\Pr[\text{unsafe delivery}] \leq k\varepsilon$ for every visiting order π and however stale $\mathcal{H}$ is. The memory-free robot satisfies the same bound, with the same k. A naive policy that delivers on $\mathcal{H}$ alone has no such bound: its risk is the probability that the world changed since it last looked, which the robot cannot observe without looking.

Algorithm 1 One mission: verify, look again, search or stop, remember

Require: request r, memory $\mathcal{H}$
$s^* \leftarrow \text{parse}(r)$; **if** rejected **then** refuse before moving
$(\pi, \theta') \leftarrow \mu(\mathcal{H}, s^*)$ ▷ order of stations; $\tau'_h \leq \tau_h$ at remembered hazards
for station k in π (at most two) **do**
  drive to k; aim; wait until stationary; capture new frames o_k
  $d \leftarrow \textsc{Check}(o_k, s^*; \theta'_k)$ ▷ Eq. (1); and the second checker, if used
  if $d =$ hazard **then**
    hold zero 2.5 s; report; **stop**
  if $d =$ approve **then**
    for side v in order(θ'_k) **do**
      move 0.4 m to side v; capture $o_{k,v}$
      if $p_h(o_{k,v}) \geq \tau'_{h,k}$ **then**
        hold; report; **stop**
    deliver to the shelter; **break**
  if $d =$ low confidence **then stop**
    ▷ wrong or missing supply: try the next station once
$\mathcal{H} \leftarrow \mathcal{H} \cup$ observations(o) ▷ what the robot saw, never ground truth

Two caveats keep this honest. First, ε is a property of the check on the views it meets; a different route meets different views, so the bound transfers only to the extent that ε is uniform over stations. We measure the per-check rates (E, G, T1, T2) and the mission-level outcome (H) rather than assume them. Second, the corollary bounds risk, not time: a stale route proposal costs a detour, which campaign H and the change-rate model measure.

Algorithm 1 summarises one mission. A *naive* memory is not verification-preserving. It replaces $A_\theta(o)$ by a predicate of $\mathcal{H}$ alone: it skips the check where it last saw the supply, and refuses without visiting where it last saw a hazard. We implement it only as a baseline. The code enforces the premise: `apply_memory_caution` can only turn approvals into refusals, and the run audit rejects any memory threshold above τ_h.

IV. Evaluation Methodology

Development data. Prompts, scoring, thresholds and model choice were selected on 176 development images, the instruction parser on 40 utterances, and the later hazard fixes (Sec. V-C) also on two 300-image hazard sets, DH and DH2 (composition: Supp. Sec. S1).

Held-out test sets. *Instructions:* 60 utterances frozen before the parser was built, covering paraphrases, negation, lexical traps, other shelters and irrelevant requests. *Images (T1):* 60 Gazebo worlds × 2 stations × 5 viewpoints, with continuous random lighting, supplies moved by up to ±0.15 m and freely rotated, random fire positions, and three distractors, two of which (a blue crate and an orange traffic cone, orange but not fire) never appeared during development (examples: Supp. Fig. S1). Labels come from the generator, never from a model. Whether the fire is visible is measured, not assumed: each view of a fire world is rendered a second time with only flames and smoke drawn in flat magenta, so occlusion

by the supply or the platform is taken into account. A view is *hidden* with at most 20 fire pixels and *ambiguous* with 21–613; the 1 ambiguous image(s) are excluded from the primary endpoints, as registered. Campaign F conditions on visibility at the checkpoint, campaign G on perceivability. These labels use no model output and are fixed before any trial.

Closed-loop campaigns. Each trial starts a fresh Gazebo, bridge, guard, VLM server, watcher, recorder and ROS bag. Campaign D (42 trials, v3.4 with $\tau_h = 0.5$) used the three controlled scenes plus fire, dim light and a box. Campaign E (60 trials, v3.5) gave every trial its own randomised world, registered before any E trial was run, in the strata nominal, swapped, empty, fire at the requested station, and fire at the *other* station, where the search itself must stop. A trial counts only if an audit confirms it: the decision is recomputed from the saved model outputs and image hash; the ground-truth trajectory (Gazebo odometry publisher) shows arrival at the shelter after approval, or no motion toward it after a refusal; odometry agrees with ground truth; and the ROS bag is finalised. Only stack failures *before* any VLM decision may be re-run once. All attempts stay in the ledger.

Metrics and statistics. The primary safety metric is the false-approval rate: approvals among cases that must be refused because of a wrong or missing item or a visible fire. It is paired with the correct-approval rate. We report Wilson 95% intervals [65]; images of one world share lighting, props and fire, so pooled rates also get a bootstrap interval that resamples whole worlds. Paired comparisons use exact McNemar tests [66] and, for times, the exact Wilcoxon signed-rank test. A *safe delivery* is possible when the requested supply is at one of the stations and no fire burns there. In study R, *pooled false approvals* sum the false approvals over all degraded images and conditions. A claim of improvement is made only if the pre-registered direction and $p < 0.05$ both hold; otherwise we write "not significant" and report the counts.

Model selection and registration audit. Qwen3.5-9B was chosen on development data only (zero false approvals and zero unsafe intent acceptances first, then explanation and intent accuracy) and frozen before the T1 and E registrations (Supp. Sec. S2). For every study with results, the registration commit precedes the results commit (holds), except campaign C (Supp. Sec. S9, Table S7).

V. Results

A. Held-out image test (T1)

Table II shows the registered system and the alternative models on T1. With the registered v3.4 rule ($\tau_h = 0.5$), Qwen3.5-9B approved 5 of 408 must-refuse images (1.2%, 95% CI [0.5, 2.8]%) and 187 of 191 must-approve images (97.9%, [94.7, 99.2]%). Its hazard question flagged 67/76 images with a visible fire. The pre-specified v3.5 rule ($\tau_h = 0.05$) gave 3 false approvals ([0.2, 2.1]%) and 187 correct approvals (97.9%), with hazard recall 69/76 (false alarms: Supp. Sec. S7). *Where the errors are.* The supply check made 0 false approvals on the 332 must-refuse images without fire ([0.0, 1.1]%), including swapped and empty stations and unseen crates. All registered false approvals came from the 48 images with the correct supply and a visible fire, 5 of which were approved: the fire was partly hidden behind the supply, with only smoke or a flame edge in view, and detection depends on the visible fire area. With the v3.5 rule, the hazard question flagged 68/69 visible fires of at least 10 000 pixels but only 1/7 smaller ones (together the 69/76 above; the 10 000-pixel cut was chosen post hoc). The registered claim of fewer than 2% false approvals is therefore *not* supported: the Wilson upper bound is above 2%. A bootstrap that resamples the 60 worlds (five views share lighting, props and fire) gives [0.2, 2.8]% for v3.4 and [0.0, 1.7]% for v3.5. Supp. Table S2 isolates the parts of the rule: removing the hazard question raises false approvals to 47 of 408, while the permuted second question and the confidence threshold matter only for weaker models (option-letter bias; Supp. Sec. S4). *Free-form generation.* Free JSON (A5) gave no parse errors with Qwen3.5-9B and the same 5 false approvals, but only 158 correct approvals against 187 with scoring. All models missed largely the same small fires (hazard recall 59 to 68 of the 76 visible fires); Supp. Fig. S2 shows descriptive risk–coverage curves over τ_s.

Is a generative VLM needed? (study Z, pre-registered). We ran contrastive zero-shot baselines with SigLIP [74] and CLIP [75], with the same decision structure and thresholds chosen on development data only (Table III; Supp. Sec. S7). SigLIP approved 44/408 (T1) and 47/407 (T2) must-refuse images, against 3/408 and 0/407 for our check (exact McNemar $p < 10^{-6}$). CLIP refused almost everything: it gave 2/191 and 0/193 correct approvals.

Conformal τ_h (study K, pre-registered, offline). Split-conformal calibration of τ_h on 211 development fires (miss rate $\leq \alpha$ with probability 0.9) gave $\tau_h = 0.0050$ ($\alpha = 0.05$) and 0.0113 ($\alpha = 0.10$). On the 76 visible T2 fires the bound held at $\alpha = 0.05$ and held at 0.10 (0/76 and 0/76 missed), with 26/193 and 4/193 false alarms on 193 fire-free must-approve images, against 0/76 and 0/193 for the frozen $\tau_h = 0.05$. Exchangeability is broken by design (small fires: 27% of calibration, 7% of T2 fires); the frozen threshold was not changed.

B. Closed-loop missions

In campaign D the system behaved as expected in 42/42 trials ([91.6, 100.0]%) with 0 wrong deliveries (median decision latency 3.05 s, maximum 6.64 s; 2 pre-decision infrastructure retries). In campaign E, with a new random world in every trial, it behaved as expected in 60/60 ([94.0, 100.0]%) with 0 wrong deliveries ([0.0, 6.0]%). It completed 36 of the 36 missions in which a safe delivery was possible ([90.4, 100.0]%); per-stratum results and trajectories are in Supp. Table S3 and Fig. S3. *Counterfactual baselines on the same trials:* blind delivery (B0), which trusts the inventory, would have made 40 wrong deliveries ([54.1, 77.3]%) and completed only 20 of 36 correctly. Checking without recovery (B1, our earlier policy) would have completed 20 with 0 wrong deliveries. *Time.* In the 34 delivered missions without search, the check

TABLE I

ALL PRE-REGISTERED STUDIES. EACH WAS REGISTERED (PROTOCOL, CONFIGURATION, ENDPOINTS AND ANALYSIS CODE) BEFORE ITS FIRST TRIAL OR INFERENCE; STUDIES WERE DESIGNED SEQUENTIALLY, EACH AFTER THE PREVIOUS RESULTS, SO REGISTRATION IS CONFIRMATORY WITHIN A STUDY (SUPP. TABLE S7). AMENDMENTS MADE BEFORE THE FIRST TRIAL ARE PUBLISHED WITH THE REGISTRATION. CLAIM: STATUS OF THE REGISTERED CLAIM (DESCRIPTIVE: NONE REGISTERED; PENDING: NOT YET RUN).

Study	Question	Data	Arms (systems)	Primary endpoint	Claim
D	Does the v3.4 system behave as expected?	42 closed-loop trials, 3 scenes + variants	1 (v3.4)	expected behaviour	descriptive
T1	Does the check generalise to unseen variation?	600 held-out images, 60 worlds	1 (v3.4; v3.5, ablations secondary)	false approvals $< 2\%$	not supported
E	Closed loop in freshly randomised worlds	60 trials, one world each	1 (v3.5; counterfactuals)	expected behaviour, wrong deliveries	descriptive
T2	Do hazard fixes chosen on dev data hold?	600 new held-out images	2 (v3.5; selected)	false approvals	not supported
F	Do they help in closed loop on small hidden fires?	24 worlds, paired	2 (v3.5, v3.6)	deliveries despite a visible fire	not supported
G	Does looking from the side help?	24 new worlds, paired	2 (v3.6, v3.6 + look-around)	deliveries despite a perceivable fire	supported
R	Is the check fail-closed under camera degradation?	300 T2 images $\times$ 12 degradations	1 (v3.6)	pooled false approvals $< 2\%$	supported
H	Can memory speed missions up without new risk?	16 sequences $\times$ 3 missions, changing worlds	3 (v3.6 with no, naive, verified memory)	time; unsafe deliveries	supported
J	Does the two-model design run on the robot?	24 worlds, paired	2 (v3.6; Qwen3.5-4B + Clef-Flash)	completes on one GPU; unsafe deliveries	not supported
V	Does the repeated-mission model predict closed-loop behaviour?	10 sequences $\times$ 6 missions, random change	3 (v3.6 with no, naive, verified memory)	prediction intervals; unsafe deliveries	pending
W	How often is a small, partly hidden fire missed, and do prior mechanisms match ours?	150 fire + 75 fire-free worlds $\times$ 3 views (image level); T1 + T2 vs. vote and conformal baselines	2 (front only; look-around), 3 baselines	look-around miss bound $\leq 15\%$ (Wilson); false approvals	pending
Z	Is a generative VLM needed?	T1 + T2, contrastive zero-shot baselines	3	false and correct approvals	supported
K	Does a conformal τ_h keep a miss bound?	211 dev fires $\rightarrow$ T2 (offline)	2 ($\alpha = 0.05, 0.10$) vs frozen	held-out fire miss rate $\leq \alpha$	supported
P	Does the hazard question transfer to real photos?	400 D-Fire test images (CC0)	1 (v3.6 hazard question)	recall; false alarms (descriptive)	descriptive
P2	Does a site-calibrated τ_h make the hazard question usable on real photos?	D-Fire: 400 calibration, 400 test images	2 wordings vs frozen	recall ≥ 0.90 and false alarms ≤ 0.20	pending

TABLE II

T1 HELD-OUT TEST, REGISTERED V3.4 RULE (EQ. 1) UNLESS MARKED FREE JSON; 408 MUST-REFUSE AND 191 MUST-APPROVE IMAGES.

Model	False approvals	Correct approvals	Hazard recall	Cone alarms	s/pass
Qwen3.5-9B	5/408	187/191	67/76	0/25	0.98
Qwen3.5-4B	9/408	190/191	59/76	0/25	1.06
Qwen3-VL-8B	4/408	183/191	68/76	0/25	0.46
Qwen2.5-VL-3B	4/408	187/191	68/76	0/25	0.83
Qwen3.5-9B (free JSON)	5/408	158/191	–	–	–
Qwen2.5-VL-3B (free JSON)	0/408	57/191	–	–	–

TABLE III

STUDY Z: ZERO-SHOT CONTRASTIVE BASELINES VERSUS OUR CHECK ON THE HELD-OUT SETS (FALSE APPROVALS / CORRECT APPROVALS).

System	T1 FA	T1 CA	T2 FA	T2 CA
Ours (T1 v3.5, T2 v3.6)	3/408 (0.7%) [0.2, 2.1]	187/191 (97.9%) [94.7, 99.2]	0/407 (0.0%) [0.0, 0.9]	190/193 (98.4%) [95.5, 99.5]
SigLIP (so400m, zero-shot)	44/408 (10.8%) [8.1, 14.2]	191/191 (100.0%) [98.0, 100.0]	47/407 (11.5%) [8.8, 15.0]	193/193 (100.0%) [98.0, 100.0]
CLIP (ViT-L/14, zero-shot)	0/408 (0.0%) [0.0, 0.9]	2/191 (1.0%) [0.3, 3.7]	0/407 (0.0%) [0.0, 0.9]	0/193 (0.0%) [0.0, 1.9]

takes about 5.5 s (scored VLM queries: 3.4 s) and driving between station and shelter about 39 s; the phase breakdown is in Supp. Table S4.

C. Fixing the remaining weakness

T1 left one failure mode: fires only partly visible behind the supply. Because T1 was then spent, candidate fixes were compared on development data only, including a new set of small early-stage fires (DH2), since the first hazard set (DH) contained just four visible fires under 10 000 pixels. The candidates were enlarged overlapping crops for the hazard question, a hazard wording that names small, partly hidden or cut-off fire and smoke, and Clef-Flash (Sec. III-C). The wording together with the crops is v3.6. Supp. Table S6 shows the results. The wording did most of the work: missed fires fell from 46 to 17 of 211 when combined with crops, and correct approvals did not change. Clef-Flash refused half of the correct deliveries when asked the original second-checker questions, which concern the safety of a *passage*; with task-worded questions it reached 97%, and agreement with Qwen gave the fewest false approvals, but the two models need 37 GB together, which does not fit on one V100.

Held-out confirmation (T2, pre-registered, 600 new images). The new draw happened to be easy: v3.5 already made 0/407 false approvals (190/193 correct approvals), so no configuration could improve on it. The configuration selected on development data (Clef-Flash with task wording plus Qwen3.5-9B v3.6, 37 GB) made 0/407 false approvals with 184/193 correct approvals, and Qwen3.5-9B v3.6 alone made 0/407 with 190/193. Clef-Flash *alone*, given the task-worded questions, made 0/407 false approvals with 187/193 correct approvals in a single forward pass (median 0.6 s per image, against 3.4 s for the scored Qwen queries; Supp. Table S4).

Closed loop: small fires (campaign F, pre-registered, paired). Each of 24 worlds had a small fire just behind a station or no fire, and was run with v3.5 and with v3.6. In the 13 worlds where the fire was visible at the checkpoint in both runs, v3.5 delivered despite the fire 12 times and v3.6 7 times (exact McNemar $p = 0.0625$, not significant; 5 discordant worlds favoured v3.6 and 0 the other). Neither system stopped falsely in the fire-free runs (v3.5 0/11, v3.6 0/11); with so few runs this says little. Six of v3.6's seven remaining misses were small fires whose hazard probabilities overlap those of fire-free views (Supp. Sec. S7).

Looking from the side (Sec. III-B, Fig. 3). Mask renders show that the same fires cover 4 600–22 000 pixels from 0.4 m to the left or right of the checkpoint. Campaign G (pre-registered, paired, new worlds) compared v3.6 with v3.6 plus look-around. In the 16 worlds whose fire was perceivable from at least one view, v3.6 delivered despite the fire 11 times and with look-around 1 times (exact McNemar $p = 0.0019$; 10 discordant worlds favoured look-around and 0 the other arm). False hazard stops in fire-free runs were 0/8 without and 0/8 with look-around. In these runs the median mission time rose from 90 s to 122 s (+32 s); the medians over all delivered runs, 102 s and 122 s, are not comparable (Supp. Sec. S7).

Study W (pre-registered, image level, **TBD** perceivable small fires) replays the check offline: the front view alone missed **TBD**, look-around **TBD** (**TBD**, 95% CI **TBD**), at **TBD** extra false stops on fire-free twins (bound **TBD**; cost **TBD**). With the same full-image input, a paraphrase vote and two conformal sets approved **TBD**, **TBD** and **TBD** must-refuse images against **TBD** for ours (distinguished from **TBD** of 3).

D. Camera degradations (study R)

Gazebo renders are noise-free and well exposed. To probe this part of the sim-to-real gap [60], we degraded the checkpoint frames of 300 held-out T2 images (30 worlds) in 12 pre-registered ways. There are six families, in the spirit of ImageNet-C [56]: sensor noise, motion blur, defocus, under- and over-exposure, and JPEG compression, each at a moderate and a severe level (Fig. 4). A frame that the robot's own

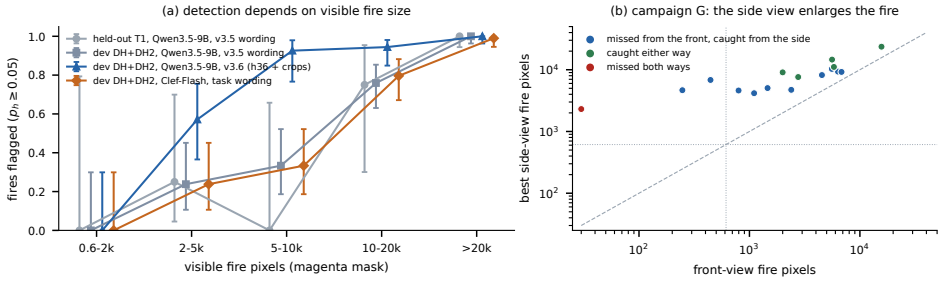


Fig. 3. (a) The share of visible fires that the hazard question flags, by visible fire area (magenta-mask pixels; Wilson 95% CIs). Fires under 2 000 pixels are missed by every configuration. The v3.6 wording with crops lifts the 2 000–10 000-pixel range. (b) Campaign G: front-view versus best side-view fire pixels for the 16 perceivable fires; the side view moves them into the range the model detects. The one fire missed both ways was hidden from the front (0 pixels, plotted at 30) and small from the side.

acquisition gate rejects (black-out, white-out or flat frame) never reaches the VLM and counts as a refusal. Supp. Table S5 gives every condition. Pooled over all conditions, the check made 0/2304 false approvals, so the registered fail-closed claim is supported. Pooling repeats the same 192 must-refuse images, so the image-level bound is per condition: at most 0/192 false approvals, Wilson upper bound 1.96%. 0 conditions increased false approvals significantly. Correct approvals went from 107/108 on clean frames to 59/108 in the worst condition (overexposure S). The gate refused 28 frames in total, and 0 of 27 visible fires were missed in any condition. Under shift, refusal rather than unsafe approval is the intended failure direction of a selective classifier [10].

Real photographs (study P). We also asked the frozen v3.6 hazard question about 400 real photographs from the test split of D-Fire [76], a CC0 fire and smoke dataset (100 each with no fire, smoke only, fire only, and both). It flagged 298/300 of the photographs with fire or smoke (95% CI [97.6, 99.8]%) and 64/100 of the fire-free ones ([54, 73]%); the false alarms concentrate on hard negatives such as hazy hillsides and sunset clouds (Supp. Sec. S7). The probabilities still rank the photographs well (AUROC 0.96, not registered), so the problem is the threshold, not the question. The question thus errs strongly toward stopping on real images, which is the safe direction but would cost availability outdoors, so a site-specific threshold calibration is a deployment prerequisite. Study P2 tested this: it chose wording and τ_h on 400 separate calibration photographs (recall ≥ 0.95 first, then the lower false-alarm rate), with wording **TBD** and $\tau_h =$ **TBD**, and applied it to 400 test photographs. It flagged **TBD** of those with fire or smoke (**TBD**) and **TBD** of the fire-free ones (**TBD**), against **TBD** and **TBD** for the frozen $\tau_h = 0.05$. The registered claim (test recall ≥ 0.90 and false-alarm rate ≤ 0.20) was **TBD**.

E. Cross-mission memory (campaign H)

Campaign H has 16 sequences of three missions to the same shelter. Lighting and prop placement are re-drawn each mission, and four scripted stories change the world between missions: the supplies are swapped and stay swapped (*relocation persists*) or are swapped back (*relocation changes*); a small fire appears and persists; a fire disappears and a new one appears. The first mission seeds all three arms; missions 2 and 3 run with no memory (M0, v3.6), naive memory (M1) and verified memory (M2). Table IV (per-mission outcomes and times: Supp. Fig. S4) summarises the 32 paired missions. On the 8 missions whose supplies stayed relocated, the median mission time was 79 s for M2 against 114 s for M0 (exact Wilcoxon $p = 0.0078$). M1 made 20 unsafe deliveries, against 0 for M2 (exact McNemar $p = 1.9 \times 10^{-6}$) and 1 for M0. Missions of one sequence share a world and a memory, so we also test at the sequence level: M1 made at least one unsafe delivery in 12 of 16 sequences and M2 in 0 (exact McNemar $p = 0.0005$). The M1–M2 contrast is large by design, since the naive arm skips checks; the comparison that bounds our risk is M2 versus M0, 0/32 versus 1/32 (Wilson upper bounds 11% and 16%), so P3 (M2 $\leq$ M0) is met as registered but informative only down to those bounds. False refusals (stops at a station whose fire was already out) were 0 for M0, 4 for M1 and 0 for M2. The registered headline claim, verified memory keeping memory's speed-up without its safety cost, is supported. M1's unsafe deliveries were 12 wrong supplies, relocated since it last looked, and 8 deliveries into a fire that had appeared since; every one of them followed a skipped check. M2 made neither kind of error and no false refusal, and it was faster in 8/8 persistent-relocation pairs. A stale proposal has a price: when the supplies had been moved back, M2 first visited the station memory proposed and then searched, and its median mission time was 113 s against 97 s for M0. M0's single unsafe delivery shows what caution is for. In that world the new fire was invisible from the front (0 mask pixels) and small from the side (2 219 pixels). Without memory, the left view scored $p_h = 0.035$, below $\tau_h = 0.05$, and the robot delivered. Memory had recorded a fire at this station two missions earlier, so M2 checked with $\tau_h^{\text{mem}} = 0.01$; its left view scored $p_h = 0.049$ and the robot stopped (Supp. Fig. S5). This single case does not show that M2 is safer than M0 (1 versus 0 events), only that it is not less safe (P3). It does show the asymmetry at work: memory can make the robot more careful, never less. Caution is local: a global $\tau_h = 0.01$ would refuse correct deliveries on fire-free views (on T2, the conformal threshold $\tau = 0.0113$ already raised false alarms to 4/193, study K), whereas memory applies it only at stations with a recent hazard. Its cost is not established: caution applied



Fig. 4. Study R: one held-out frame with a visible fire under the six registered degradations (severe level; the moderate level is milder). Over-exposure washes out the flames in this example. Frames that the acquisition gate rejects count as refusals.

TABLE IV
CAMPAIGN H, MISSIONS 2–3 (PAIRED). UNSAFE = WRONG SUPPLY OR A PERCEIVABLE FIRE AT THE DELIVERING STATION. TIME: MEDIAN OVER ALL 32 MISSIONS.

Arm	Unsafe	Wrong supply	Into fire	False refusals	Median time [s]
M0 no memory (v3.6)	1/32	0	1	0	83
M1 naive memory	20/32	12	8	4	28
M2 verified memory (ours)	0/32	0	0	0	73

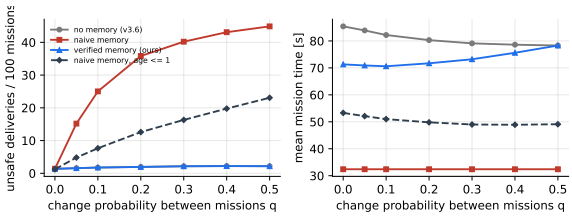


Fig. 5. Pre-registered model of repeated missions, from the per-check rates measured in G and H: unsafe deliveries and mean mission time as the world changes more often, for no memory, naive memory, a recency-gated naive memory that trusts only the previous mission (M1age), and verified memory (a model prediction, not an observation). Recency gating reduces but does not remove the risk of trusting memory; verified memory keeps the memory-free risk.

in 12 of the 32 M2 missions, only 4 of them at a fire-free station, with 0 false refusals there (Wilson upper bound 49%). A global $\tau = 0.0113$ would also have stopped M0 in Q09.3 (descriptive, not registered; Supp. Sec. S7). Every paired mission is listed in Supp. Table S8. Fig. 5 extends the campaign. At a change probability of $q = 0.2$ per mission, the model predicts 35.83 unsafe deliveries per 100 missions for naive memory, against 1.95 for verified memory and 1.95 without memory; the mean mission times are 32.4, 71.7 and 80.3 s. Keeping every check has a price: verified memory keeps 27% of naive memory's modelled time saving at $q = 0$ (14.1 of 53.0 s) and 0% at $q = 0.5$. The saving is bounded by the arena: with two stations memory can only avoid one search detour ($\delta = 30$ s). With k stations and a supply that relocates uniformly with probability q, verified memory saves $\delta\,((k-1) - qk)/2$ per mission (9 s at $k = 2$ and 80 s at $k = 8$ for $q = 0.2$; break-even at $q^* = (k-1)/k$; Supp. Sec. S7) while every delivering station is still checked, so the benefit grows with the site and the risk bound does not change.

Study V tested the model in closed loop (10 sequences of 6 missions, supply location and fire toggling at random with $q = 0.2$, predictions registered before the first trial). The registered 95% interval for the paired saving M0−M2 was **TBD** s (median **TBD** s); we observed **TBD** s. Unsafe deliveries were **TBD** (M0), **TBD** (M1) and **TBD** (M2), against registered intervals **TBD**, **TBD** and **TBD**. The claim that the model predicts the closed loop is **TBD**; that verified memory keeps its speed-up without a safety cost under random change, **TBD** (M1 vs. M2 by sequence: exact McNemar $p =$ **TBD**).

F. Two models on one GPU (campaign J)

On T2, Clef-Flash together with Qwen3.5-4B, the pair that fits on one 32 GB V100, made 0/407 false approvals (185/193 correct). Campaign J ran this pair on the robot, paired with v3.6 on 24 worlds with swapped, empty and small-fire stations; it behaved as expected in 23/24 (v3.6: 24/24, exact McNemar $p = 1.0000$), with unsafe deliveries: 1 (v3.6: 0; Wilson upper bounds 20% and 14%); stack failures: 0. Peak GPU memory was 28.0 GB, and the median check latency was 5.1 s for the two-model system and 5.7 s for v3.6; on our host, alternating ≈46 GB of weights thrashed the page cache, so both include model I/O (registered deviation note). The registered claim (two-model design on the simulated robot, no extra risk) is not supported.

Instruction parsing. On 60 held-out utterances the hybrid parser was correct for 55/60 with 0 unsafe acceptances (TF-IDF parser: 31/60, 2 unsafe); all five errors were refusals, so they fail closed.

VI. DISCUSSION

When should a robot trust its memory? Skipping a recent successful check is efficient, and its risk grows with how often the world changes, which the robot cannot observe without looking (campaign H, Fig. 5; registered headline claim: supported). Observation 1 and Corollary 1 make the question moot for safety: memory may decide the order and strictness of the checks, never their existence, and buys time in persistent worlds at the price of a detour when the world changes back.

Moving beats thinking harder. The largest single safety gain came from moving 0.4 m (G), not from a larger model, a better prompt or a second model (T2, whose fires were easy, cannot rank models on safety). A hidden fire carries no pixels to reason about.

What pre-registration bought us. Three registered claims failed or were not significant: false approvals below 2% on T1, a closed-loop benefit of better wording and crops (F), and any detectable change on T2 (too easy a draw). The first two exposed the hidden-fire weakness and led to the look-around; unregistered, each could have been rationalised away. A check takes about 5.5 s and the look-around about 32 s, against about 39 s of driving between station and shelter, which is favourable where a wrong delivery costs a round trip.

VII. Limitations

All results are from simulation: simple meshes, two supply types, fixed station geometry, one robot, a stylised fire, and sensor degradations rather than the appearance gap of real objects or smoke. Campaign H scripts its changes, assumes independent changes, spans three missions and two stations, and does not compare memory caution with a global threshold at equal false-stop rate (Supp. Sec. S7). Corollary 1 assumes a per-check false-approval bound uniform over the stations a route meets. Counterfactual baselines were not run physically, barely visible fires (21–613 pixels) are excluded, and the guard is hand-written, not formally synthesised [31], [32]. The next step is a real platform. Threats to validity: Supp. Sec. S9.

VIII. Conclusion

Three lessons carry beyond our arena. First, an off-the-shelf local VLM can gate a physical action with few false approvals if it is only asked to score closed-set answers, and if motion stays behind an independent guard (T1: 3 of 408 must-refuse images for v3.5; 0/2304 under 12 camera degradations). Second, the failures that remained were failures of viewpoint, not of the model: a fire hidden behind the supply was missed from the front and found from the side (G: 11 to 1 of 16). Third, memory should change *where* and *how carefully* a robot looks, never *whether* it looks; under this rule a robot gains speed from what it remembers while its risk bound stays that of a robot without memory (H: supported). The two-model design is a registered claim reported in Sec. V-F (J: not supported). All of this is simulation; the photo study shows that a site-specific threshold is the first thing a real deployment must calibrate. Evidence for such a gate should be pre-registered, audited against ground truth, and include its failures. We release code, worlds, test sets, protocols, ledgers and per-trial evidence; every number here is produced from `experiments/results` by `scripts/paper_numbers.py`.

References

[1] B. Balcik, B. M. Beamon, and K. Smilowitz, "Last mile distribution in humanitarian relief," *Journal of Intelligent Transportation Systems*, vol. 12, no. 2, pp. 51–63, 2008. [Online]. Available: https://doi.org/10.1080/15472450802023329

[2] R. R. Murphy, *Disaster Robotics*. Cambridge, MA: The MIT Press, 2014. [Online]. Available: https://doi.org/10.7551/mitpress/9407.001.0001

[3] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing robot experiences for failure explanation and correction," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3468–3484. [Online]. Available: https://proceedings.mlr.press/v229/liu23g.html

[4] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang *et al.*, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/70a06501001e1820fd1eb9ee821302d2-Abstract-Conference.html

[5] C. Agia, R. Sinha, J. Yang, Z. Cao *et al.*, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 689–723, coRL 2024 (PMLR volume dated 2025); method name: Sentinel. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[6] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang, "Large language models are not robust multiple choice selectors," in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://openreview.net/forum?id=shr9PXz7T0

[7] P. Pezeshkpour and E. Hruschka, "Large language models sensitivity to the order of options in multiple-choice questions," in *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 2024, pp. 2006–2017. [Online]. Available: https://doi.org/10.18653/v1/2024.findings-naacl.130

[8] Y. Li, Y. Du, K. Zhou, J. Wang *et al.*, "Evaluating object hallucination in large vision-language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2023, pp. 292–305. [Online]. Available: https://doi.org/10.18653/v1/2023.emnlp-main.20

[9] Y. Sun and Y. Zhang, "When memory lies: An empirical study of spatial memory staleness in VLM agents," 2026, arXiv preprint.

[10] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NIPS)*, vol. 30, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html

[11] Cloudflare, "Clef-Flash model card," Hugging Face model repository, 2026. [Online]. Available: https://huggingface.co/Cloudflare/clef-flash

[12] D. Seto, B. Krogh, L. Sha, and A. Chutinan, "The Simplex architecture for safe online control system upgrades," in *Proceedings of the 1998 American Control Conference (ACC)*. IEEE, 1998, pp. 3504–3508. [Online]. Available: https://doi.org/10.1109/ACC.1998.703255

[13] L. Sha, "Using simplicity to control complexity," *IEEE Software*, vol. 18, no. 4, pp. 20–28, 2001. [Online]. Available: https://doi.org/10.1109/MS.2001.936213

[14] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018. [Online]. Available: https://doi.org/10.1073/pnas.1708274114

[15] J. M. Hofman, A. Chatzimparmpas, A. Sharma, D. J. Watts, and J. Hullman, "Pre-registration for predictive modeling," 2023. [Online]. Available: https://arxiv.org/abs/2311.18807

[16] Y. Du, K. Konyushkova, M. Denil, A. Raju *et al.*, "Vision-language models as success detectors," in *Proceedings of The 2nd Conference on Lifelong Learning Agents (CoLLAs)*, ser. Proceedings of Machine Learning Research, vol. 232. PMLR, 2023, pp. 120–136. [Online]. Available: https://proceedings.mlr.press/v232/du23b.html

[17] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez *et al.*, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Robotics: Science and Systems XXI (RSS)*, 2025, method name: FAIL-Detect. [Online]. Available: https://doi.org/10.15607/RSS.2025.XXI.073

[18] E. Zhou, Q. Su, C. Chi, Z. Zhang *et al.*, "Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 6919–6929. [Online]. Available: https://doi.org/10.1109/CVPR52734.2025.00649

[19] R. Sinha, A. Elhafsi, C. Agia, M. Foutter, E. Schmerling, and M. Pavone, "Real-time anomaly detection and reactive planning with large language models," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[20] A. Oloo, Z. Altaweel, Y. Hayamizu, P. Liu, Y. Ding, S. Amiri, H. Yang, A. Kaminski, C. Esselink, C. Paxton, X. Zhang, and S. Zhang, "Robot planning and situation handling with active perception," 2026, arXiv preprint; method name: VAP-TAMP.

[21] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh *et al.*, "Robots that ask for help: Uncertainty alignment for large language model planners," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 661–682. [Online]. Available: https://proceedings.mlr.press/v229/ren23a.html

[22] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021. [Online]. Available: https://arxiv.org/abs/2107.07511

[23] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.

[24] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, pp. 1641–1662, 2025.

[25] B. Ichter, A. Brohan, Y. Chebotar, C. Finn *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2022, pp. 287–318, coRL 2022 (PMLR volume dated 2023). Author order follows the PMLR record; the arXiv version (2204.01691) lists Michael Ahn first ("Ahn et al., 2022"). [Online]. Available: https://proceedings.mlr.press/v205/ichter23a.html

[26] J. Robinson, C. M. Rytting, and D. Wingate, "Leveraging large language models for multiple choice question answering," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=yKbprarjc5B

[27] S. Kadavath, T. Conerly, A. Askell, T. Henighan *et al.*, "Language models (mostly) know what they know," 2022. [Online]. Available: https://arxiv.org/abs/2207.05221

[28] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[29] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate before use: Improving few-shot performance of language models," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 12 697–12 706. [Online]. Available: https://proceedings.mlr.press/v139/zhao21c.html

[30] T. Wu, C. Zhou, G. Zhao, H. Cao, Y. Pu, and J. Yang, "When robots should say "I don't know": Benchmarking abstention in embodied question answering," 2025, arXiv preprint.

[31] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer *et al.*, "Safe reinforcement learning via shielding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11797

[32] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista *et al.*, "Control barrier functions: Theory and applications," in *18th European Control Conference (ECC)*, 2019, pp. 3420–3431. [Online]. Available: https://doi.org/10.23919/ECC.2019.8796030

[33] K.-C. Hsu, H. Hu, and J. F. Fisac, "The safety filter: A unified view of safety-critical control in autonomous systems," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 7, no. 1, pp. 47–72, 2024.

[34] A. Robey, Z. Ravichandran, V. Kumar, H. Hassani, and G. J. Pappas, "Jailbreaking LLM-controlled robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 948–11 956. [Online]. Available: https://doi.org/10.1109/ICRA55743.2025.11128119

[35] Z. Ravichandran, A. Robey, V. Kumar, G. J. Pappas, and H. Hassani, "Safety guardrails for LLM-enabled robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4649–4656, 2026. [Online]. Available: https://doi.org/10.1109/LRA.2026.3667488

[36] B. Zitkovich, T. Yu, S. Xu, P. Xu *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183, author order follows the PMLR record; the arXiv version (2307.15818) lists Anthony Brohan first ("Brohan et al., 2023"). [Online]. Available: https://proceedings.mlr.press/v229/zitkovich23a.html

[37] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao *et al.*, "OpenVLA: An open-source vision-language-action model," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 2679–2713, coRL 2024 (PMLR volume dated 2025). [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[38] A. Anwar, J. Welsh, J. Biswas, S. Pouya, and Y. Chang, "ReMEmbR: Building and reasoning over long-horizon spatio-temporal memory for robot navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 2838–2845.

[39] Z. Wang, B. Yu, J. Zhao, W. Sun, S. Hou, S. Liang, X. Hu, Y. Han, and Y. Gan, "KARMA: Augmenting embodied AI agents with long-and-short term memory systems," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[40] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Proc. Robotics: Science and Systems (RSS)*, 2022.

[41] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.

[42] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," in *Proc. Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 23–72.

[43] P. Biber and T. Duckett, "Dynamic maps for long-term operation of mobile service robots," in *Proc. Robotics: Science and Systems (RSS) I*, 2005, pp. 17–24.

[44] T. Krajník, J. P. Fentanes, J. P. S. S. Santos, and T. Duckett, "FreMEn: Frequency map enhancement for long-term mobile robot autonomy in changing environments," *IEEE Transactions on Robotics*, vol. 33, no. 4, pp. 964–977, 2017.

[45] X. Yin and H. Du, "GLOVE: Global verifier for LLM memory-environment realignment," 2026, arXiv preprint.

[46] G. Tang, Q. Jia, G. Chen, T. Li, Z. Huang, Z. Lv, and N. Ji, "VLM-DEWM: Dynamic external world model for verifiable and resilient vision-language planning in manufacturing," 2026, arXiv preprint.

[47] A. Aydemir, A. Pronobis, M. Göbelbecker, and P. Jensfelt, "Active visual object search in unknown environments using uncertain semantics," *IEEE Transactions on Robotics*, vol. 29, no. 4, pp. 986–1002, 2013.

[48] A. Wandzel, Y. Oh, M. Fishman, N. Kumar, L. L. S. Wong, and S. Tellex, "Multi-object search using object-oriented POMDPs," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2019, pp. 7194–7200.

[49] Z. Zeng, A. Röfer, and O. C. Jenkins, "Semantic linking maps for active visual object search," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020, pp. 1984–1990, arXiv:2006.10807.

[50] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang, "ESC: Exploration with soft commonsense constraints for zero-shot object navigation," in *Proc. 40th Int. Conf. Machine Learning (ICML), PMLR 202*, 2023, pp. 42 829–42 842, arXiv:2301.13166.

[51] J. Mason and B. Marthi, "An object-based semantic world model for long-term change detection and semantic querying," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 3851–3858.

[52] D. M. Rosen, J. Mason, and J. J. Leonard, "Towards lifelong feature-based mapping in semi-static environments," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2016, pp. 1063–1070.

[53] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[54] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *Autonomous Robots*, vol. 42, no. 2, pp. 177–196, 2018.

[55] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[56] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[57] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," 2019, arXiv preprint.

[58] S. Chen, J. Gu, Z. Han, Y. Ma, P. Torr, and V. Tresp, "Benchmarking robustness of adaptation methods on pre-trained vision-language models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[59] M. C. Schiappa, S. Vyas, H. Palangi, Y. S. Rawat, and V. Vineet, "Robustness analysis of video-language models against visual and language perturbations," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022.

[60] S. Höfer, K. Bekris, A. Handa, J. C. Gamboa, M. Mozifian, F. Golemo *et al.*, "Sim2Real in robotics and automation: Applications and challenges," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 2, pp. 398–400, 2021.

[61] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees *et al.*, "Evaluating real-world robot manipulation policies in simulation," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 3705–3728, coRL 2024 (PMLR volume dated 2025); method name: SIMPLER.

[62] A. Avižienis, "The N-version approach to fault-tolerant software," *IEEE Transactions on Software Engineering*, vol. SE-11, no. 12, pp. 1491–1501, 1985.

[63] J. C. Knight and N. G. Leveson, "An experimental evaluation of the assumption of independence in multiversion programming," *IEEE Transactions on Software Engineering*, vol. SE-12, no. 1, pp. 96–109, 1986.

[64] H. Kress-Gazit, K. Hashimoto, N. Kuppuswamy, P. Shah *et al.*, "Robot learning as an empirical science: Best practices for policy evaluation," 2024. [Online]. Available: https://arxiv.org/abs/2409.09491

[65] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. [Online]. Available: https://doi.org/10.1080/01621459.1927.10502953

[66] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. [Online]. Available: https://doi.org/10.1007/BF02295996

[67] J. Tobin, R. Fong, A. Ray, J. Schneider *et al.*, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30. [Online]. Available: https://doi.org/10.1109/IROS.2017.8202133

[68] S. Macenski, T. Foote, B. Gerkey, C. Lalancette, and W. Woodall, "Robot operating system 2: Design, architecture, and uses in the wild," *Science Robotics*, vol. 7, no. 66, p. eabm6074, 2022. [Online]. Available: https://doi.org/10.1126/scirobotics.abm6074

[69] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 3, 2004, pp. 2149–2154. [Online]. Available: https://doi.org/10.1109/IROS.2004.1389727

[70] Qwen Team, "Qwen3.5: Towards native multimodal agents," Official blog post and Hugging Face model card (Qwen/Qwen3.5-9B, Apache-2.0), Feb. 2026, model card: https://huggingface.co/Qwen/Qwen3.5-9B. [Online]. Available: https://qwen.ai/blog?id=qwen3.5

[71] S. Bai, K. Chen, X. Liu, J. Wang *et al.*, "Qwen2.5-VL technical report," 2025. [Online]. Available: https://arxiv.org/abs/2502.13923

[72] S. Bai, Y. Cai, R. Chen, K. Chen *et al.*, "Qwen3-VL technical report," 2025. [Online]. Available: https://arxiv.org/abs/2511.21631

[73] M. Chen, "Introducing Clef: Our open-source decision models, and new RL fine-tuning platform," Cloudflare Blog, Oct. 2026. [Online]. Available: https://blog.cloudflare.com/clef-decision-models

[74] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

[75] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[76] P. V. A. B. Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, no. 18, pp. 15 349–15 368, 2022.